

# CO2 Emission Analysis & Prediction

Upload World Bank Dataset (CSV)



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world\_bank\_dataset.csv 12.9KB



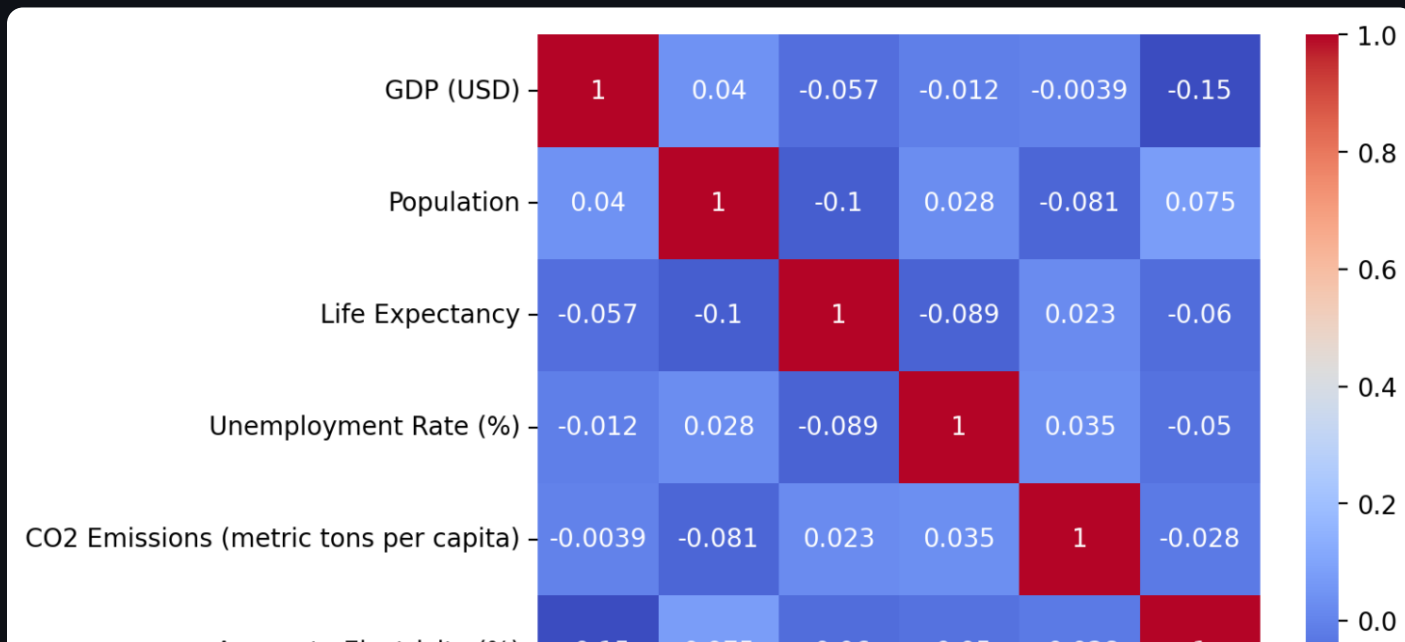
## Data Preview:

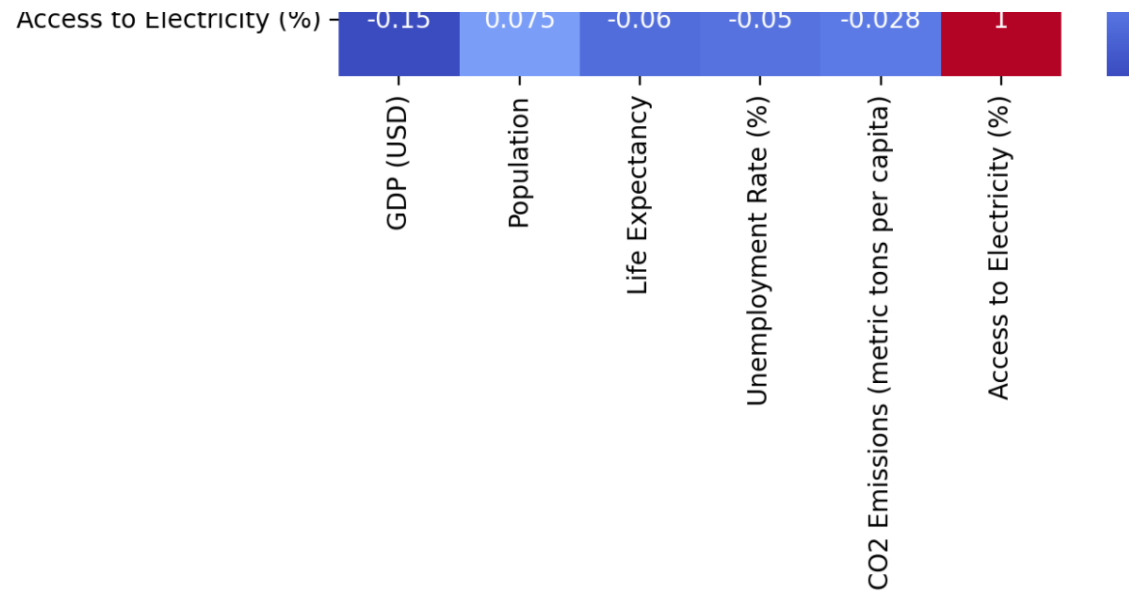
|   | Country | Year | GDP (USD)          | Population | Life Expectancy | Unemployment Rate (%) | CO2 Em |
|---|---------|------|--------------------|------------|-----------------|-----------------------|--------|
| 0 | Brazil  | 2010 | 1493220000000      | 829020000  | 66.7            | 3.81                  |        |
| 1 | Japan   | 2011 | 17562700000000     | 897010000  | 61.4            | 17.98                 |        |
| 2 | India   | 2012 | 16426880000000.004 | 669850000  | 69.1            | 16.02                 |        |
| 3 | Mexico  | 2013 | 11890010000000     | 113800000  | 80.1            | 6.26                  |        |
| 4 | India   | 2014 | 2673020000000      | 29710000   | 62.7            | 3.1                   |        |

## Summary Statistics:

|       | Year   | GDP (USD)         | Population     | Life Expectancy | Unemployment Rate (%) | CO2 Emission |
|-------|--------|-------------------|----------------|-----------------|-----------------------|--------------|
| count | 200    | 200               | 200            | 200             | 200                   |              |
| mean  | 2014.5 | 10568674550000    | 738790850      | 66.3245         | 13.2717               |              |
| std   | 2.8795 | 5547703222772.701 | 438995561.0858 | 9.8189          | 6.8042                |              |
| min   | 2010   | 1011720000000     | 9970000        | 50.5            | 2.27                  |              |
| 25%   | 2012   | 5774120000000     | 353377500      | 57.775          | 6.81                  |              |
| 50%   | 2014.5 | 10506150000000    | 721670000      | 64.9            | 13.47                 |              |
| 75%   | 2017   | 15034512500000    | 1090860000     | 74.5            | 18.5425               |              |
| max   | 2019   | 19983770000000    | 1498060000     | 84.9            | 24.79                 |              |

## Feature Correlation Heatmap:





## K-Means Clustering

Clusters Assigned to Data:

|   | Country | Cluster |
|---|---------|---------|
| 0 | Brazil  | 4       |
| 1 | Japan   | 1       |
| 2 | India   | 0       |
| 3 | Mexico  | 2       |
| 4 | India   | 2       |

## Life Expectancy Prediction using Random Forest

Random Forest RMSE: 3.759928780575504

## Predict CO2 Emissions

GDP (USD)

0.04

- +

Access to Electricity (%)

0.00

- +

Industry Output

0.00

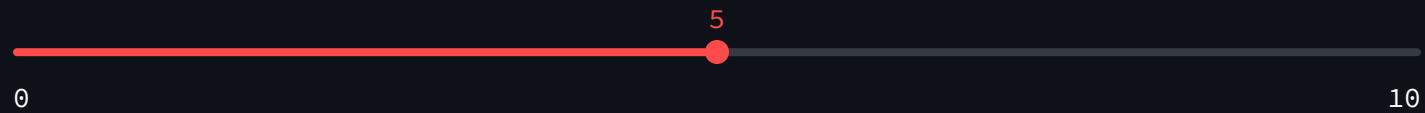
- +

Population

0.00

- +

Regulation Level



Predict CO2 Emissions

